

MODULE - 1

- *. Explain AC load line for VDB Amp.
- *. Explain the design constraints of VDB arrangement with circuit & relevant equations. How does R_E provide a negative feedback action to stabilize bias current.
- *. Starting from fundamentals define h parameters & obtain an h parameter eqt ckt of CE configuration.
- *. Develop h parameter model for Transistor Amplifier hence draw h parameter model for CB, CE & CC.
- *. Derive an expression for voltage gain A_v of small signal CE BJT Amplifier.
- *. Write a neat diagram & waveform explain CC Amplifier & its AC emitter resistance, voltage gain & I/P, O/P impedance of the base. (***)
- *. Write a neat circuit diagram & explain TSEB of BJT.

MODULE - 2

- *. Explain with neat circuit diagram the MOSFET drain to gate feedback resistor biasing.
- *. Explain with neat ckt diagram MOSFET ckt using fixed V_G .
- *. Derive the expression for characterizing parameters of CS MOSFET Amp without source resistor using hybrid π eqt ckt.
- *. Write a note on 3 basic configurations of MOSFET Amp.
- *. Derive an expression for g_m & A_v for the MOSFET Amp ckt.
- *. Draw the small signal eqt ckt model for MOSFET & obtain expression for A_v (voltage gain).
- *. What is transconductance of MOSFET & mention 3 different expressions used to calculate transconductance.
- *. Explain the high frequency response of a CB Amp using MOSFET & Derive upper cut off frequency.

MODULE - 3

- *. Derive an expression for output of an Inverting summing amplifier with 3 i/p's & hence prove ckt can act averaging amp (****)
- *. Explain the working of Instrumentation amp. Mention its Application. (***)
- *. Explain working of Inverting Schmitt trigger & derive equation for trigger points.
- *. State Ideal characters of op-amp (***)
- *. Briefly explain zero crossing detector.
- *. Explain the working of pulse width modulator using IC-555 with waveforms (***)
- *. Explain monostable operation of IC 555 timer with neat diagram (***)
- *. with a neat diagram explain RC phase shift Oscillator. (***)
- *. Explain operation of Wein bridge Oscillator. Derive its frequency of oscillation. (***)
- *. Explain Hartley Oscillator

MODULE - 4

- *. Draw block diagram of current voltage series feedback Amplifier & derive an expression for i/p resistance, A_v & o/p resistance. (12 marks) (***)
- *. Explain series shunt feedback Amplifier. Determine i/p & o/p resistance of the amp. (***)
- *. Explain gain de-sensitivity & effect on bandwidth with application of negative feedback in Amplifier. (12 marks)
- *. Explain properties of negative feedback.
- *. Draw general feedback structure & explain necessary expressions
- *. Explain IC18 Amplifier with relevant equations
- *. Explain non Inverting first Order low pass filter with unity gain
- *. Explain VC15 with relevant equations.

MODULE -5.

- *. Draw ckt diagram of class B push pull Amp & explain operation with relevant equaⁿ's. (***)
- *. Explain transformer coupled class A Amplifier also derive expression R_L .
- *. Derive expression for I/p resistance for a shunt feedback Amp.
- *. How power Amplifiers are classified explain them briefly
- *. with a ckt diagram, waveforms & equations explain SCR phase control. (***)
- *. Explain bidirectional Thyristors. (***)

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